

FLARE : Technology Stack & Architecture

Flight Airfare Real-Time Economic Index Platform • Smart India Hackathon (SIH) Technical Briefing

Executive Presentation Pitch (30-Second Summary for Judges):
"FLARE is built on a decoupled, high-performance architecture combining **Next.js 16** and **FastAPI**. It ingests live airfares across 8+ OTAs (MakeMyTrip, Goibibo, EaseMyTrip, IndiGo Direct) via **Firecrawl**, **Scrape.do**, and **Apify**. Data is sanitized using **Median Absolute Deviation (MAD)** outlier detection, split into itemized components (**Base Rate**, **Tax & GST**, and **Convenience Fee**), and modeled into a Laspeyres-weighted airfare index cross-verified with **MoSPI Government CPI** benchmarks—complete with interactive **Leaflet GIS route corridors** and real-time **Recharts** financial analytics."

1. Frontend & User Experience (UX / UI)

Technology	Role in Architecture	Engineering Highlights
Next.js 16 (App Router)	Core Web Framework & SSR	Server Components, dynamic client routing (/search, /explorer, /economic), Turbopack compilation
React 19 & TypeScript 5	UI Rendering & Type Safety	Strict contracts, Suspense boundary streaming, zero runtime type errors, interactive state hooks
Tailwind CSS v4	Design System & Theming	Glassmorphic cards, dynamic light/dark modes, customized HSL color tokens, micro-animations
Leaflet & React-Leaflet	Interactive GIS Route Network	Custom HTML divIcon pulsing nodes, great-circle flight corridor polylines, map-to-search redirection
Recharts (v3)	Financial Data Visualizations	Stacked bar charts (Base vs. Tax vs. Fee), CPI area trends, carrier price distributions
Date-fns	Temporal Date Engine	Advance booking days, ISO time serialization, flight duration calculation
Cinematic Intro	Brand & 3D Experience	High-framerate video splash with smooth canvas transition into live dashboard

2. Backend Microservices & API Architecture

Technology	Role in Architecture	Engineering Highlights
FastAPI	Asynchronous REST Framework	Async non-blocking I/O endpoints, BackgroundTasks for live scraping, auto OpenAPI/Swagger docs
Python 3.12+ / 3.14	Core Computing Language	Async event loop (asyncio), robust math operations, vectorized statistical aggregations
Uvicorn (ASGI)	Application Server	Lightning-fast ASGI production server handling high concurrent requests without thread locks
SQLAlchemy 2.0 (ORM)	Database Relational Mapping	Declarative models, connection pooling, indexed multi-column filter queries
Alembic	Schema Migrations	Version-controlled database migrations with repeatable upgrade/downgrade scripts
SQLite / PostgreSQL	Relational Database Engine	Stores 1,000+ flight observations, airport metadata, and official MoSPI CPI historical series
Pydantic v2	Data Validation & Schemas	Strict input/output contracts, ISO-8601 parsing, itemized fare breakdown validation

3. Multi-Source Scraping & Data Ingestion Pipeline

Service / Tool	Target Platform	Ingestion Capabilities
Firecrawl API	Government Portals & Booking Engines	AI-driven clean markdown conversion, DOM extraction of dynamic JS content
Scrape.do API	Indian OTAs & Direct Airlines	Proxy rotation, anti-bot bypass, Indian residential IPs (geoCode=IN), full JS rendering
Apify Google Flights Actor	Google Flights Matrix	Bulk airline flight card extraction, multi-segment route scraping
Amadeus GDS API	Official Aviation Schedules	GDS flight timetable validation, schedule consistency checks
BeautifulSoup4 (BS4)	HTML Parsing Engine	Fast DOM traversal, regex pattern extraction for INR fares, carrier codes, and flight timings
HTTPX (Async)	HTTP/2 Network Client	Concurrent parallel collectors across 8+ OTAs (MMT, Goibibo, EaseMyTrip, Yatra, Ixigo, IndiGo, Air India)

4. Economic & Statistical Airfare Index Modeling

Model / Algorithm	Economic Purpose	Mathematical Implementation
Laspeyres & Jevons Index	Weighted Inflation Metric	Weighted arithmetic & geometric means measuring basket price changes against base periods
Fare Component Breakdown	Transparent Pricing Yields	Strict decomposition: Total Fare = Base Rate + Tax & GST + Portal Convenience Fee
MoSPI CPI Integration	Govt Benchmark Calibration	Incorporates official data.gov.in Transport weights (1.86% national airfare weighting)
Median Absolute Deviation (MAD)	Outlier Elimination	Robust statistical filtering detecting promotional flash sales and extreme holiday spikes
Haversine Geodesic Distance	Route Yield Normalization	Spherical distance calculation (lat/lng) to estimate per-passenger-km fare yields
Availability Pressure Index	Supply / Demand Indicator	Ratio of sold-out vs. available booking classes across advance purchase windows

5. DevOps, Reporting & Version Control

Technology	Functionality	Production Relevance
ReportLab (Python)	Automated PDF Generation	Renders official methodology and technical presentation documentation programmatically
Docker & Docker Compose	Containerized Orchestration	Multi-container setup (docker-compose.yml) encapsulating frontend, backend, and database
Git & GitHub	Version Control & Security	Clean repository with GitHub Push Protection compliance and sanitized secrets